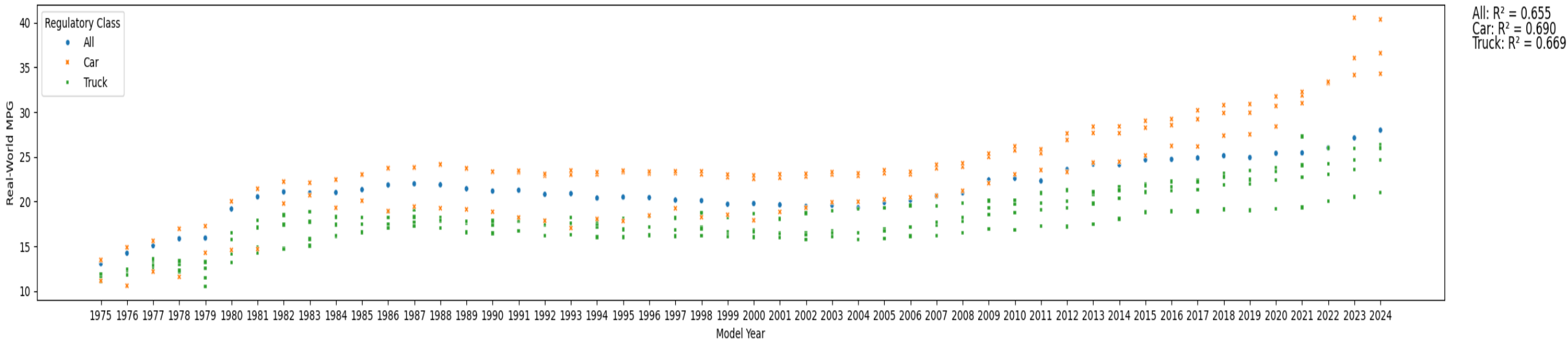


Exploring the DATASET



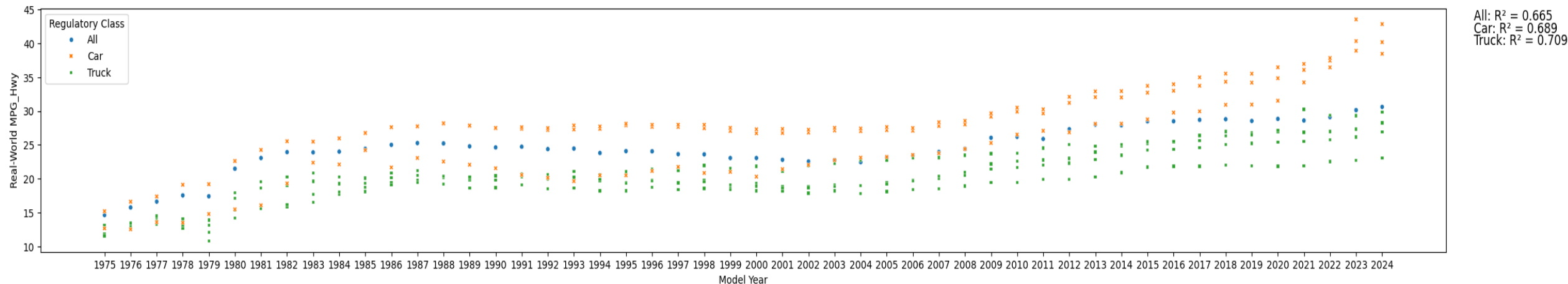
About the Dataset

- Source: EPA Automotive Trends Report
- Years covered: 1975 – 2024
- Key features: Model year, Vehicle Weight, City MPG, Highway MPG, Regulatory Class
- Outcome variable: Real-World MPG
- This dataset currently has 401 rows and 13 columns describing vehicle characteristics and performance metrics



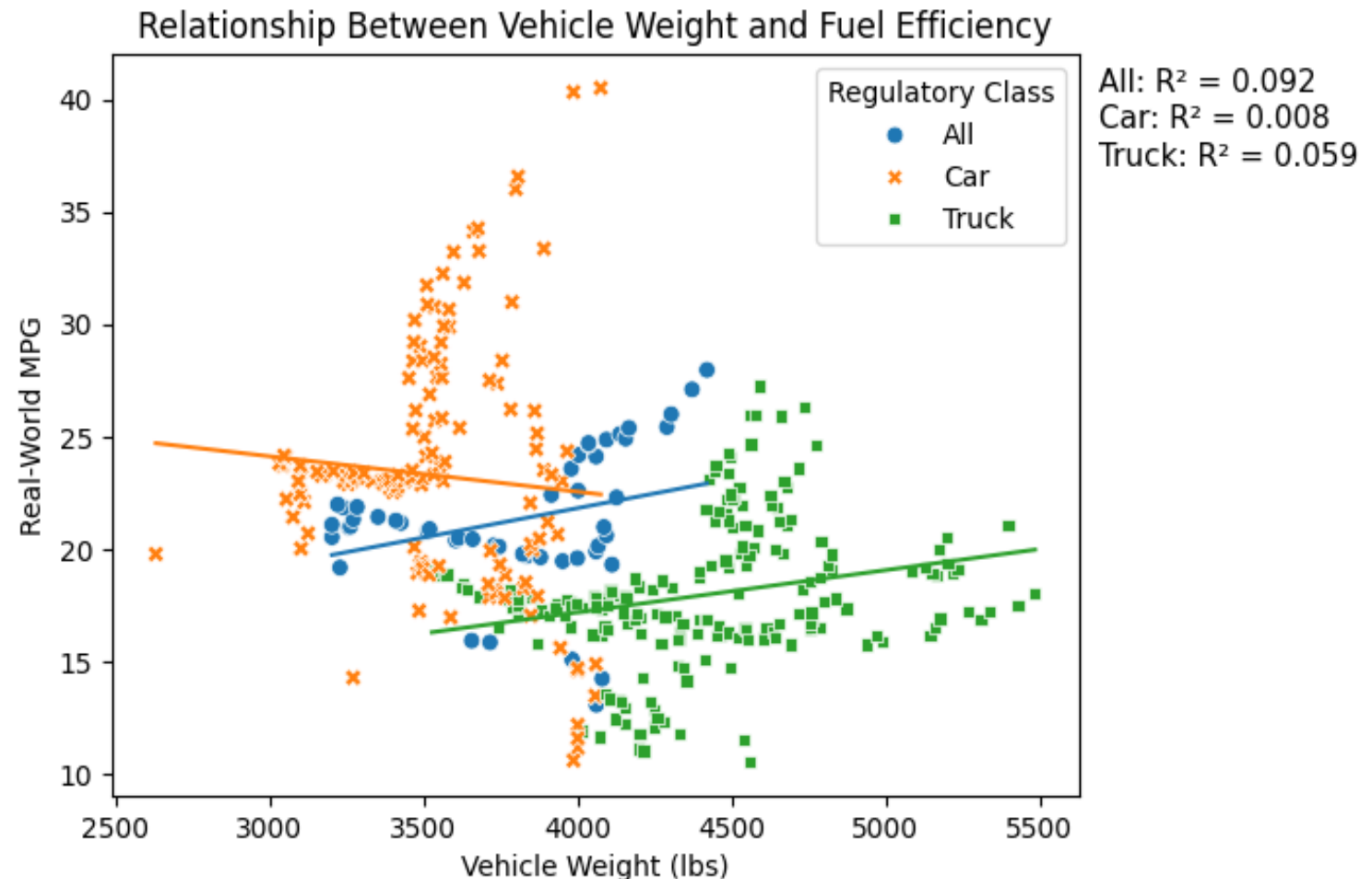
IS THERE A CORRELATION BETWEEN YEAR PRODUCED AND FUEL EFFICIENCY?

- MPG has increased steadily across all vehicle regulatory classes from 1975 – 2024
- Cars consistently show higher MPG than truck across all years
- Greater variance in recent years suggests growing vehicle diversity



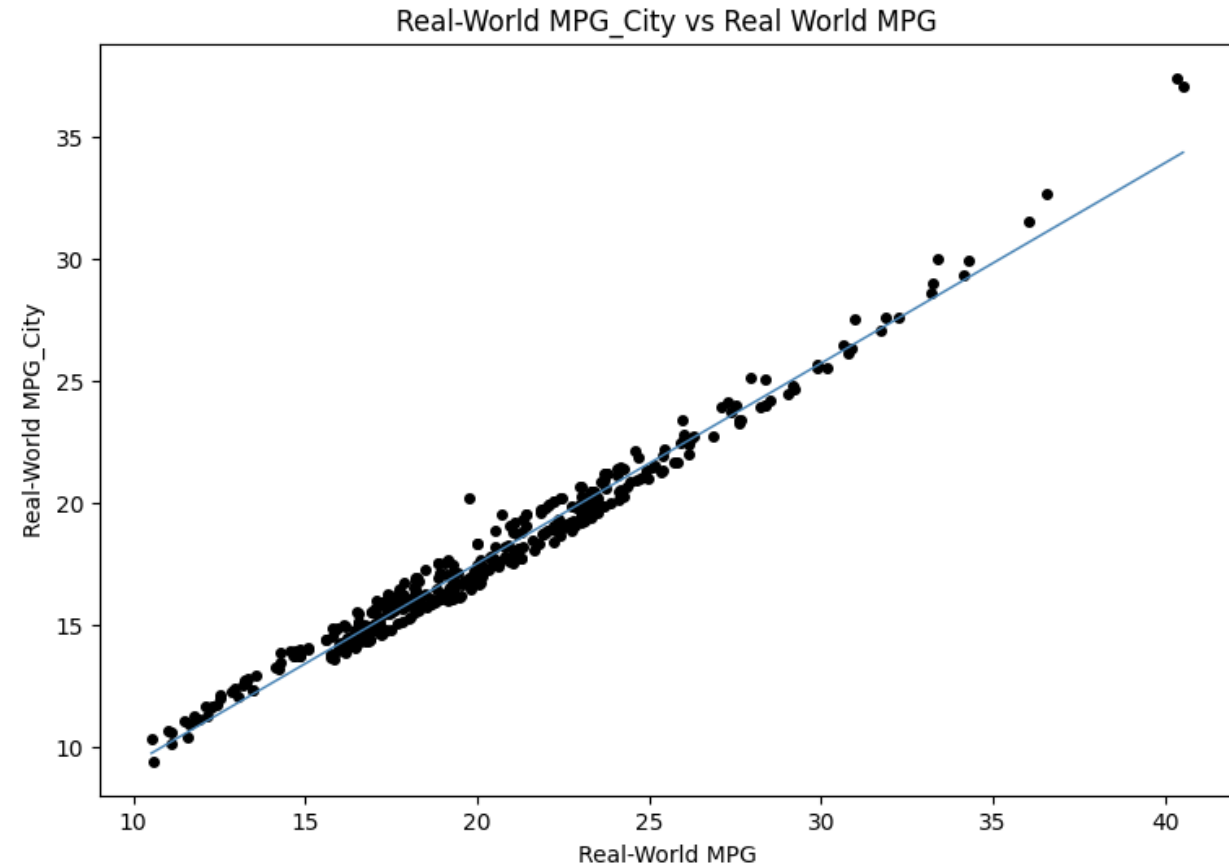
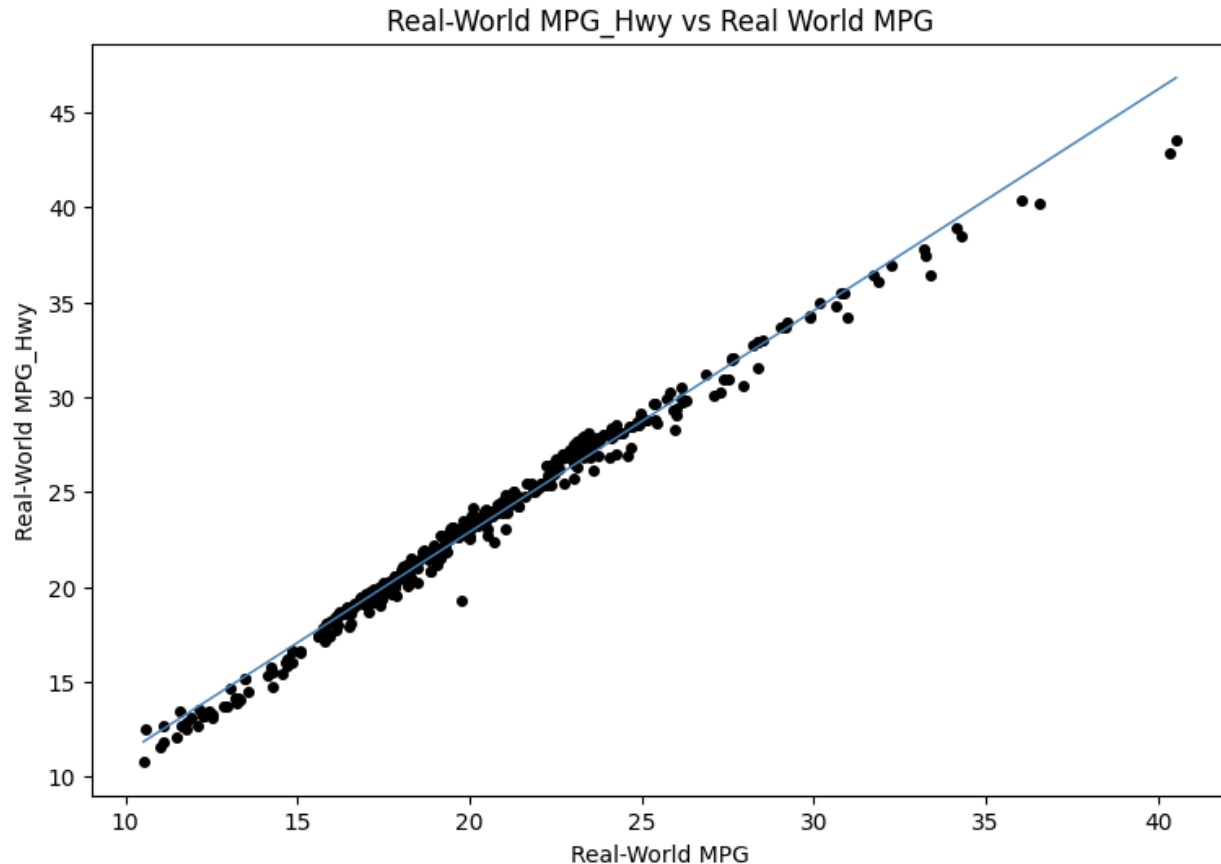
CAN WE PREDICT MPG BASED ON THE WEIGHT OF A VEHICLE?

- *Cars show more MPG variability at similar weights, suggesting that overall lighter vehicles are more efficient.*
- *Trucks cluster at higher weights with lower and more consistent MPG*
- *Wide variance across all classes suggests weight alone doesn't fully explain MPG.*

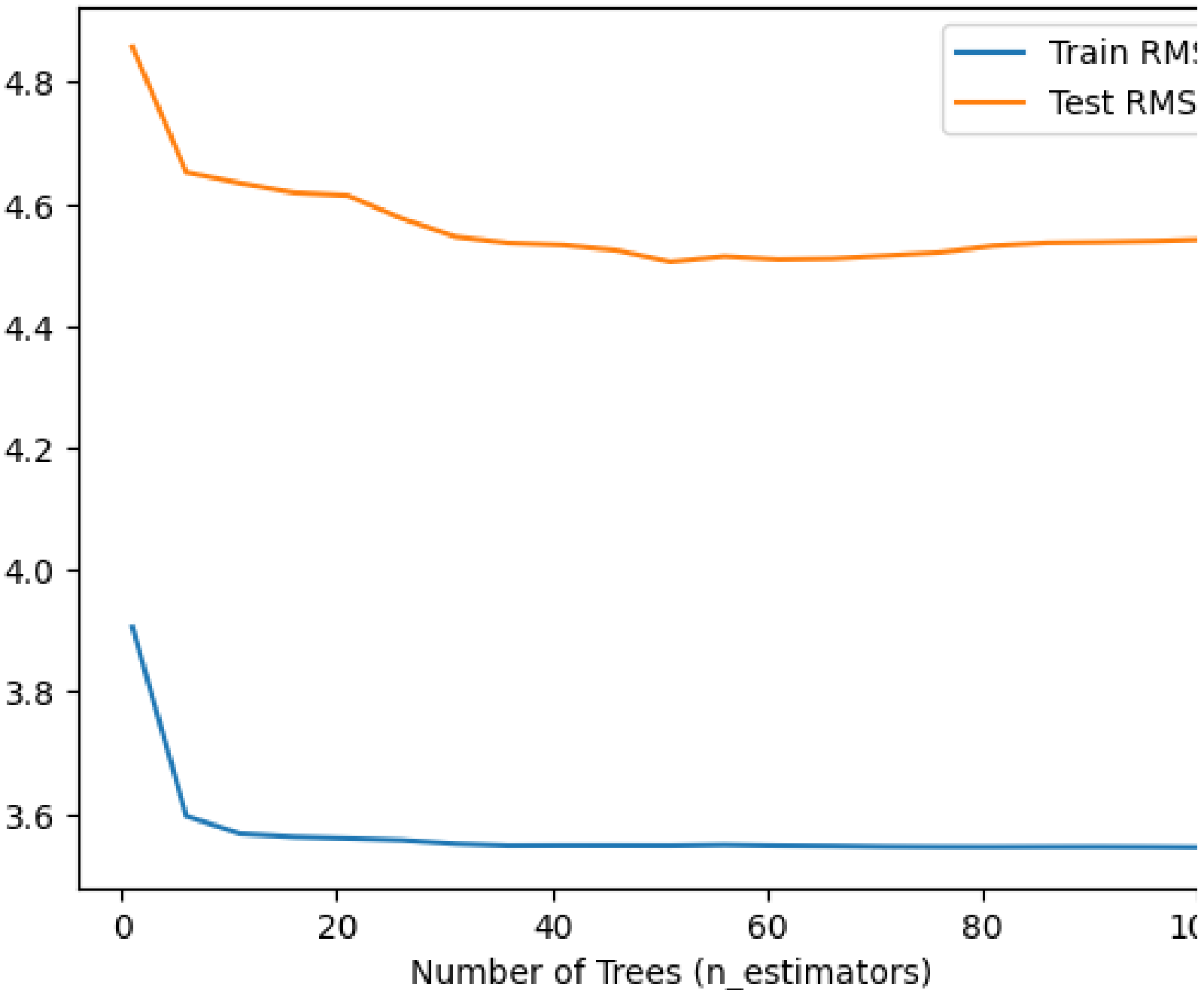


Does City or Highway MPG Better Predict Overall Mileage?

- Both city and highway MPG show strong linear relationships with overall mileage
- Highway MPG points cluster more tightly around the trend line, indicating a stronger relationship
- City MPG shows slightly more scatter, suggesting it is a weaker predictor of overall mileage.

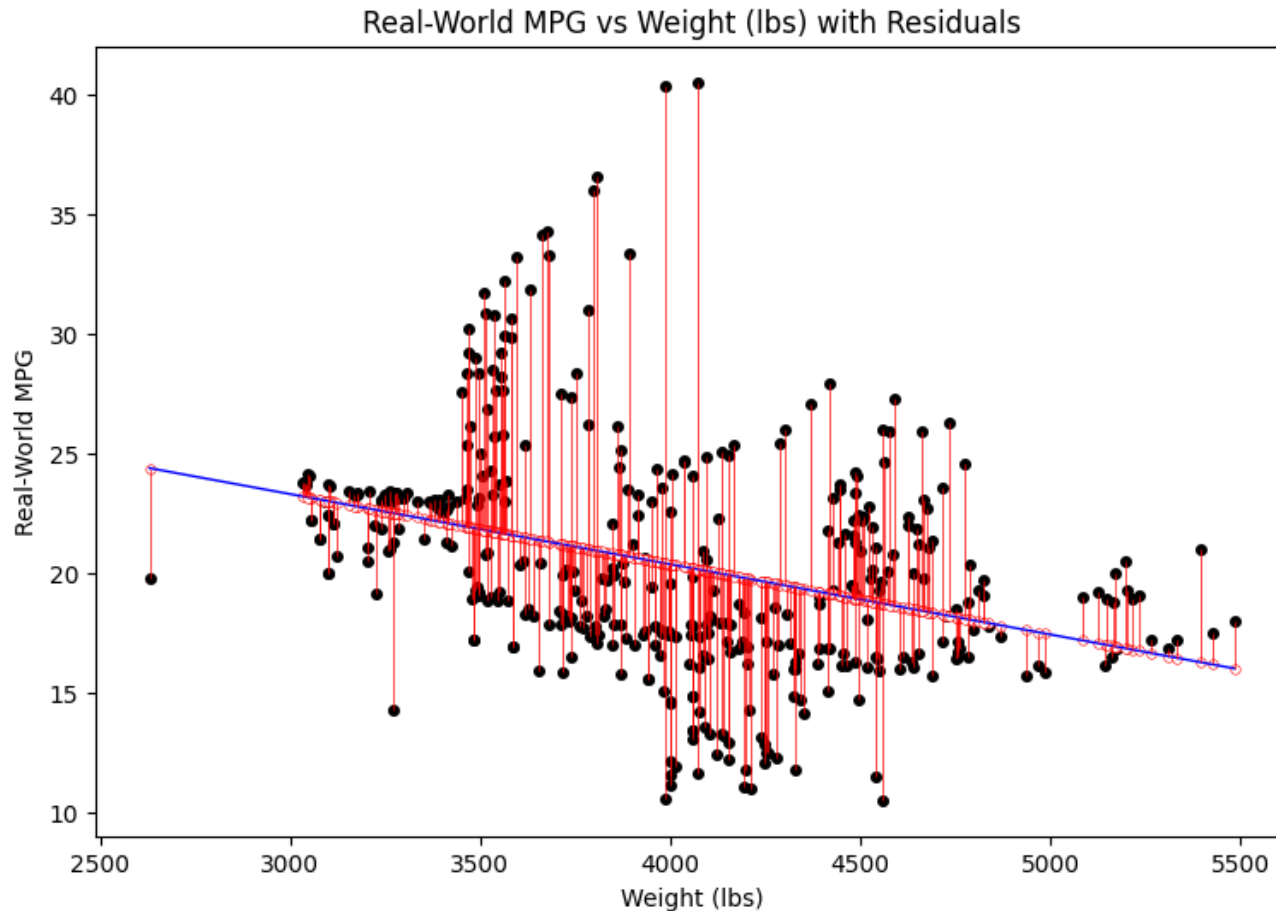


Random Forest Model Tuning



Model Results: Year vs MPG

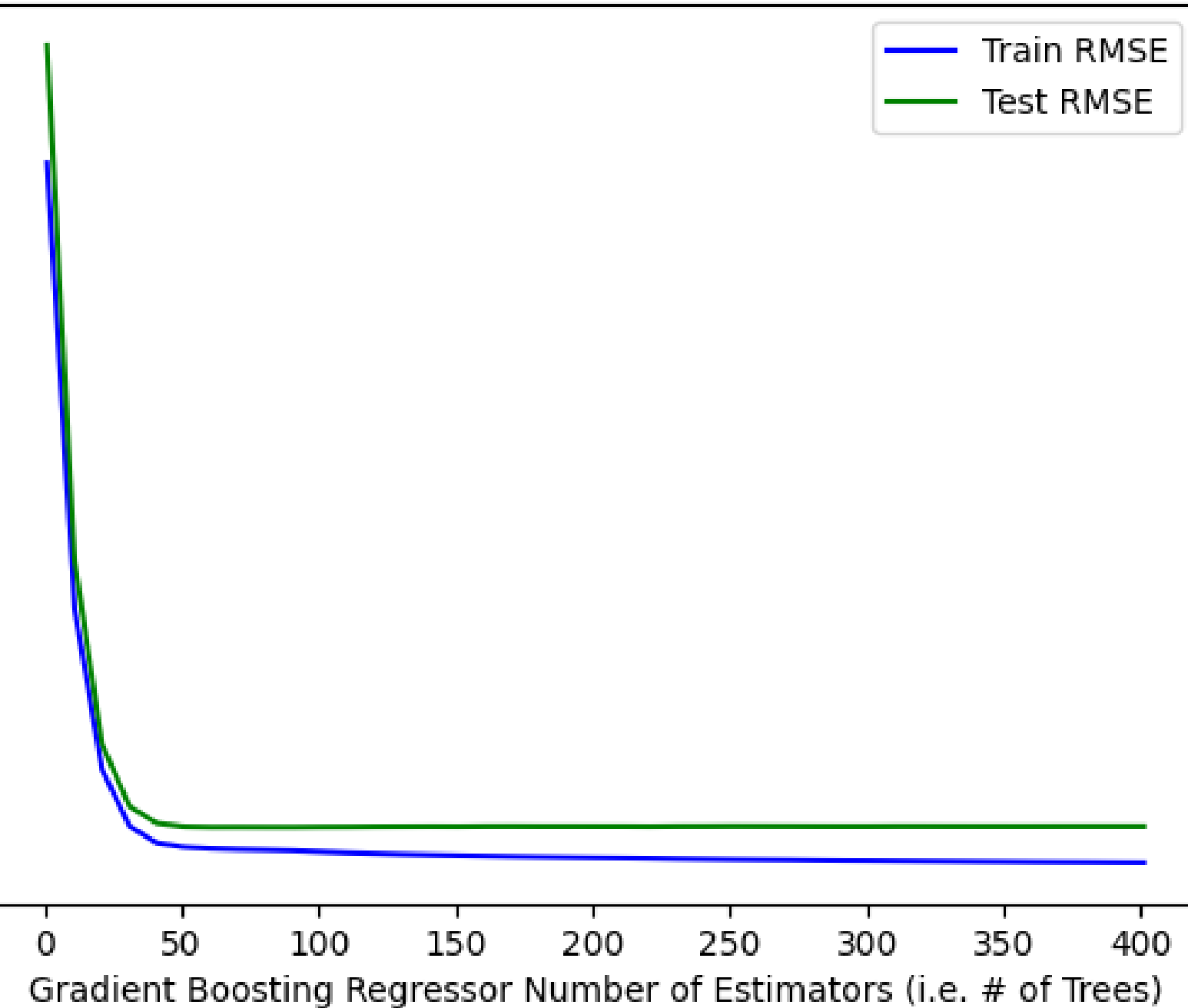
- Random Forest Model
- Test/Train split of 70/30
- No Cross Validation – train/test split was sufficient
- Evaluation Metric: RMSE
- RMSE Gap indicates mild overfitting



Can we predict mpg based on the weight of a vehicle?

- Linear Regression Model
- Test/Train split of 80/20
- No Cross Validation – train/test split was sufficient
- Evaluation Metric: R-Squared Value
- Evaluation Metric: $R^2 = 0.059$.

Gradient Boosting Regressor Model Tuning



Does higher city mpg translate to higher mileage overall or does higher highway mpg translate to higher mileage values overall?

- Gradient Boosting Regressor Model
- Test/Train split of 70/30
- No Cross Validation – train/test split was sufficient
- Evaluation Metric: Feature Importance
- Highway MPG feature importance: 74%

Conclusion

Is there a correlation between year produced and fuel efficiency?

Reject H_0 — Model Year has a measurable relationship with fuel efficiency, but the random forest's overfitting suggests it is a weak predictor alone; additional features would strengthen the model.

H_0 : There is no correlation

H_1 : There is a correlation

Can we predict mpg based on the weight of a vehicle?

Reject H_0 — vehicle weight significantly affects MPG, but low R^2 suggests weight alone is insufficient for reliable predictions; additional features are needed..

H_0 : There is no linear relationship between vehicle weight and MPG (slope = 0)

H_1 : There is a linear relationship between vehicle weight and MPG (slope $\neq 0$)

Conclusion

Does higher city mpg translate to higher mileage overall or does higher highway mpg translate to higher mileage values overall?

H₀: Neither city MPG nor highway MPG has a significant relationship.

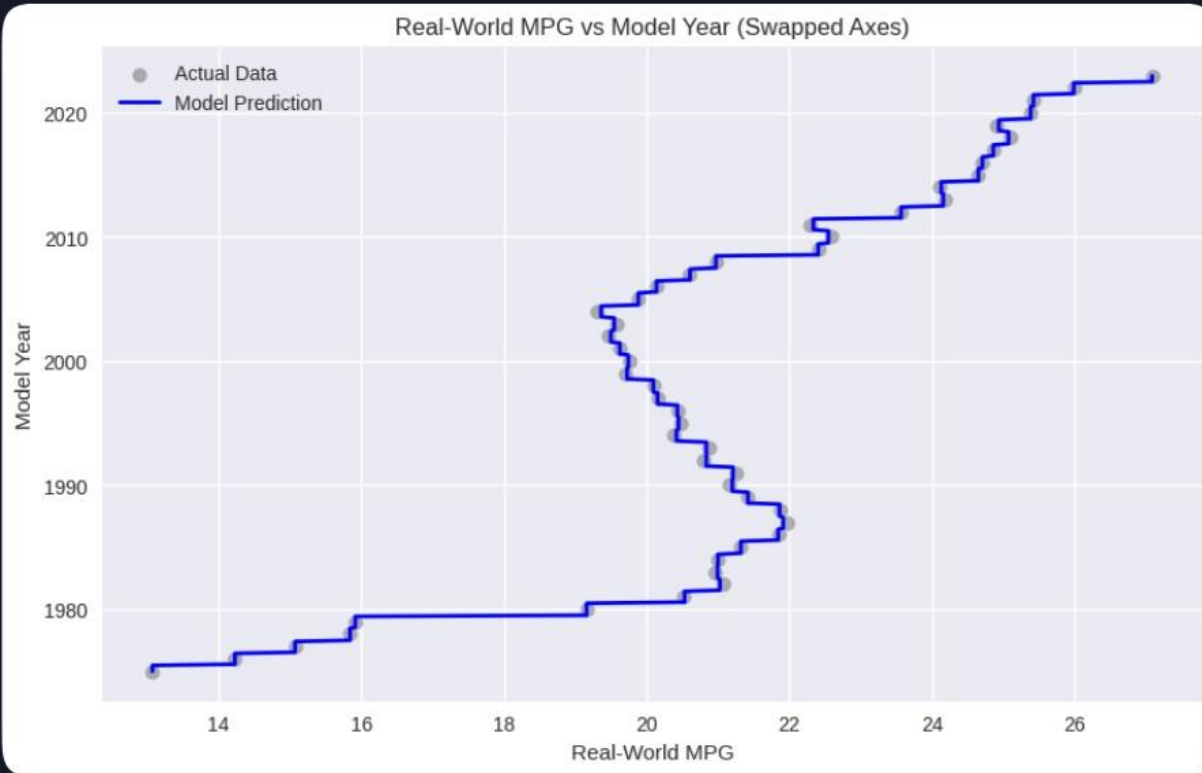
H₁: At least one of city MPG or highway MPG has a significant relationship with overall mileage.

What I might do next time:

Reject H₀ — highway MPG is the stronger predictor of overall mileage ($R^2 = 0.987$, 74% feature importance), with the model generalizing well and showing minimal overfitting.

- I would incorporate larger and more recent datasets.
- I would focus on creating a neural network to better capture the non linear relationships in the dataset.

Generated Image
copilot.com



AI Tools Used

- Open AI Chat GPT
- Microsoft Copilot

My Experience

- ChatGPT – Used to debug linear regression code and interpret model output.
- Copilot – Used to generate visualizations like the MPG vs Model Year chart shown.
- Overall, reduced time spent on syntax errors and formatting.

References:

- Microsoft. (2024). Microsoft Copilot [Large language model]. <https://copilot.microsoft.com>
- OpenAI. (2024). ChatGPT (Version 4) [Large language model]. <https://chat.openai.com>
- U.S. Environmental Protection Agency, Office of Air and Radiation, Office of Transportation and Air Quality. (2019). The EPA Automotive Trends Report: Greenhouse Gas Emissions, Fuel Economy, and Technology since 1975 [Data set]. Data.gov. <https://catalog.data.gov/dataset/the-epa-automotive-trends-report-greenhouse-gas-emissions-fuel-economy-and-technology-1975>

THANK YOU

